

Quiz (Carolina Reaper)

- Q1.** A logistics company uses the function $y = 2x$ to model the distance traveled by a ship. If the ship is delayed by 3 hours, the new function is $y = 2(x - 3)$. What type of transformation occurred?
- (A) Vertical shift
(B) Horizontal shift
(C) Reflection over x-axis
(D) Stretch
(E) Compression
- Q2.** The function $y = x^2$ models the flow of traffic in a city. If the traffic flow is reflected over the y-axis, the new function is
- (A) $y = (-x)^2$
(B) $y = x^2 + 2$
(C) $y = -x^2$
(D) $y = x^2 - 2$
(E) $y = (x + 2)^2$
- Q3.** A delivery company uses the function $y = 3x$ to model the time it takes to deliver packages. If the company hires more drivers, the new function becomes $y = 3x/2$. What type of transformation occurred?
- (A) Vertical stretch
(B) Vertical compression
(C) Horizontal stretch
(D) Horizontal compression
(E) Reflection over x-axis
- Q4.** The function $y = x^2 + 2$ models the cost of shipping goods. If the cost is doubled, the new function is
- (A) $y = 2x^2 + 2$
(B) $y = x^2 + 4$
(C) $y = 2(x^2 + 2)$
(D) $y = x^2 + 2x$
(E) $y = (x + 2)^2$
- Q5.** A transportation company uses the function $y = 2x - 1$ to model the number of passengers on a bus. If the bus is replaced by a larger bus, the new function becomes $y = 2(2x - 1)$. What type of transformation occurred?
- (A) Vertical stretch
(B) Vertical compression
(C) Horizontal stretch
(D) Horizontal compression
(E) Reflection over y-axis
- Q6.** The function $y = x^2 - 3$ models the time it takes for a package to be delivered. If the delivery time is increased by 2 hours, the new function is
- (A) $y = x^2 - 1$
(B) $y = x^2 + 2$
(C) $y = (x + 2)^2 - 3$
(D) $y = x^2 - 3 + 2$
(E) $y = x^2 - 5$
- Q7.** A shipping company uses the function $y = 3x + 2$ to model the cost of shipping goods. If the cost is reflected over the x-axis, the new function is
- (A) $y = -3x + 2$
(B) $y = 3x - 2$
(C) $y = -3x - 2$
(D) $y = 3x + 2$
(E) $y = -3x + 4$
- Q8.** The function $y = 2x^2$ models the flow of traffic in a city. If the traffic flow is compressed horizontally by a factor of 2, the new function is
- (A) $y = 2(2x)^2$
(B) $y = 2x^2/2$
(C) $y = 2(x/2)^2$
(D) $y = 8x^2$
(E) $y = x^2/2$

Open Ended Questions (FRQ)

- Q9.** A logistics company uses the function $y = x^2 - 4$ to model the cost of shipping goods. If the company wants to increase the cost by 20
- Q10.** The function $y = 3x - 2$ models the time it takes for a package to be delivered. If the delivery time is delayed by 1 hour and the new function is $y = 3(x - 1) - 2$, compare the transformed graph to the parent function and describe any differences.

Chef's Tips

- Tip 1: Provide clear instructions and examples
- Tip 2: Encourage students to use graphs and tables to visualize transformations
- Tip 3: Emphasize the importance of labeling axes and including units